Preliminary Validation & Benchmarking

CABINGUARD-ADI: A Zero-Compliance Multimodal In-Cabin Framework for Prodromal Prediction and Fail-Safe Actuation in Acute Driver Incapacitation

TSYP14 Technical Challenge — Planned Simulation, Dataset and Benchmarking Methodology

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1. Validation Objective

Phase 1 validation is designed to evaluate the feasibility of the proposed architecture before a full real-sensor vehicle implementation.

Because deliberately inducing acute syncopal arrest, tonic-clonic seizure, or severe cognitive-autonomic deterioration in live drivers would be unethical, the planned evaluation combines public de-identified baseline datasets with explicitly documented synthetic event trajectories.

The evaluation asks:

1. Can adaptive filtering improve recoverability of mechanical cardiac information under vibration?
2. Can personalized calibration reduce false alarms from inter-driver variability?
3. Can multimodal temporal reasoning separate the target etiologies from nominal and distractor behavior?
4. Can spoofed, contradictory, missing, replayed, or modified inputs be handled safely?
5. Can the implementation satisfy real-time processing requirements?

2. Baseline and Supporting Datasets

The Phase 1 evaluation assigns every dataset only to the modality or behavior it legitimately supports.

Dataset	Modality / Status	Planned Role
DMD Driver Monitoring Dataset	Optical driver monitoring	Baseline seated-driver head, gaze, and upper-body behavior.
UBFC-rPPG	Facial video / pulse	Baseline facial chrominance and pulse-processing evaluation.
IEEE DataPort SCG	Mechanical cardiac recoil	Biomechanical proxy for mechanical cardiac waveform-processing studies.
Zenodo Record 15199100	Pressure-seat matrix	Baseline CoP, lumbar-contact, and posture characterization.
In_Cabin_Behavior_Driver_safe_drive	Gated in-cabin video	Seated cabin geometry and visual-behavior baseline after access; no unverified medical labels are assumed.
Ala Mahlak Driver Monitoring Dataset	Driver-monitoring visual data	Auxiliary visual baseline; class-specific use only after verification of downloaded labels.
ADMS Dataset	Labeled driver-behavior images	Nominal/distractor support for documented behaviors including eye closure, yawning, seat-belt use, smoking, and mobile-phone use.
Radar-Based Vital Sign Monitoring	FMCW cardiorespiratory data	Auxiliary vital-sign processing reference only; not treated as an in-cabin acute-medical-event dataset.
ADL-Dataset	Pending direct content inspection	Candidate activity/posture source; no modality or label claim until verified locally.

3. Dataset Evidence Boundary

No listed public dataset is treated as direct clinical evidence for all three target acute medical events. Real samples supply baseline visual, postural, or physiological behavior according to their modality. Acute syncope, tonic-clonic seizure, and cognitive-autonomic trajectories are introduced only as explicitly marked feature-level synthetic scenarios for Phase 1 software feasibility testing.

4. Feature-Level Synthetic Event Protocol

Let $x_s^{(b)}(t)$ be a feature extracted from a real baseline sample. A controlled synthetic feature trajectory is defined as:

$$x_s^{(\mathrm{syn})}(t) = x_s^{(b)}(t) + \alpha_s g_k(t), \tag{1}$$

where α_s is the configured magnitude and $g_k(t)$ is the temporal envelope for scenario class k.

A generic prodrome-to-acute envelope is:

$$g_k(t) = \begin{cases} 0, & t < t_{p,k}, \\ \dfrac{t - t_{p,k}}{t_{a,k} - t_{p,k}}, & t_{p,k} \le t < t_{a,k}, \\ 1, & t \ge t_{a,k}, \end{cases} \tag{2}$$

where $t_{p,k}$ and $t_{a,k}$ are configured prodromal and acute onsets.

When a scenario requires movement toward a defined target $x^*_{s,k}$:

$$x_s^{(\mathrm{syn})}(t) = (1 - g_k(t))x_s^{(b)}(t) + g_k(t)x^*_{s,k}. \tag{3}$$

For oscillatory motor features:

$$x_s^{(\mathrm{syn})}(t) = x_s^{(b)}(t) + A_s g_k(t)\sin(2\pi f_k t + \phi), \tag{4}$$

using only the previously defined CABINGUARD-ADI ranges 3–6 Hz for clonic seizure motion and 4–8 Hz for cognitive-autonomic tremor.

For missing/degraded sources:

$$x_s^{(\mathrm{deg})}(t) = m_s(t)x_s^{(b)}(t) + (1 - m_s(t))\eta_s(t), \tag{5}$$

with $m_s(t) \in \{0, 1\}$ denoting source availability.

The four planned scenario families remain Syncope, Tonic-Clonic Seizure, Cognitive-Autonomic Deficit, and Distractors/Spoofing/Faults. These operators create **simulation inputs**, not clinical observations.

Each derived sample retains provenance:

$$\mathcal{R} = \{\text{dataset}, \text{source sample}, \text{real/synthetic}, \text{scenario}, \text{phase}, \text{injected features}\}.$$

Synthetic derivatives from one source subject/session must remain in the same data split as the original source to prevent train/test leakage.

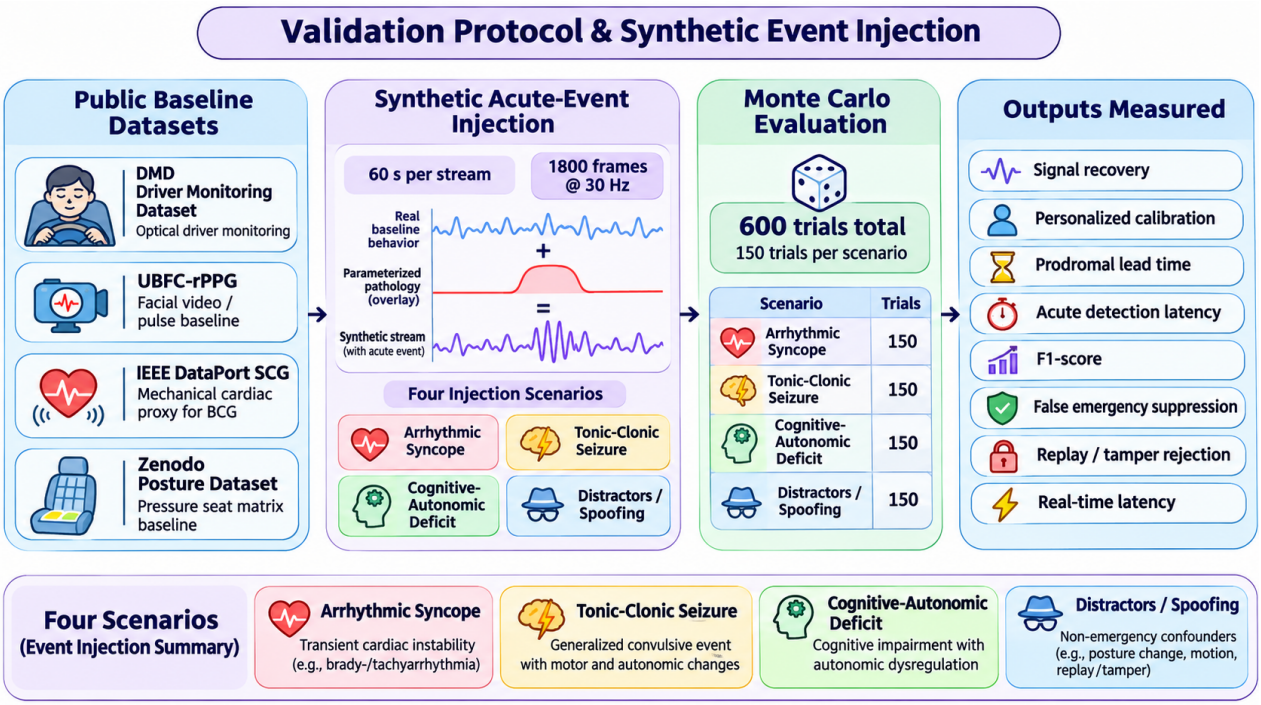


Figure 1: Original validation-protocol infographic retained unchanged. Numerical trial counts shown inside the legacy artwork are treated as planned test configuration, not completed experimental evidence.

5. Signal-Conditioning Benchmark

The Butterworth + NLMS stage will be evaluated by comparing signal quality before and after adaptive vibration cancellation:

$$\Delta\text{SNR} = \text{SNR}_{\text{after}} - \text{SNR}_{\text{before}}. \tag{6}$$

Mechanical heart-rate error will be measured as:

$$\text{MAE}_{HR} = \frac{1}{N}\sum_{i=1}^{N} |\widehat{HR}_i - HR_i|. \tag{7}$$

No SNR gain or MAE improvement is claimed before the benchmark is executed.

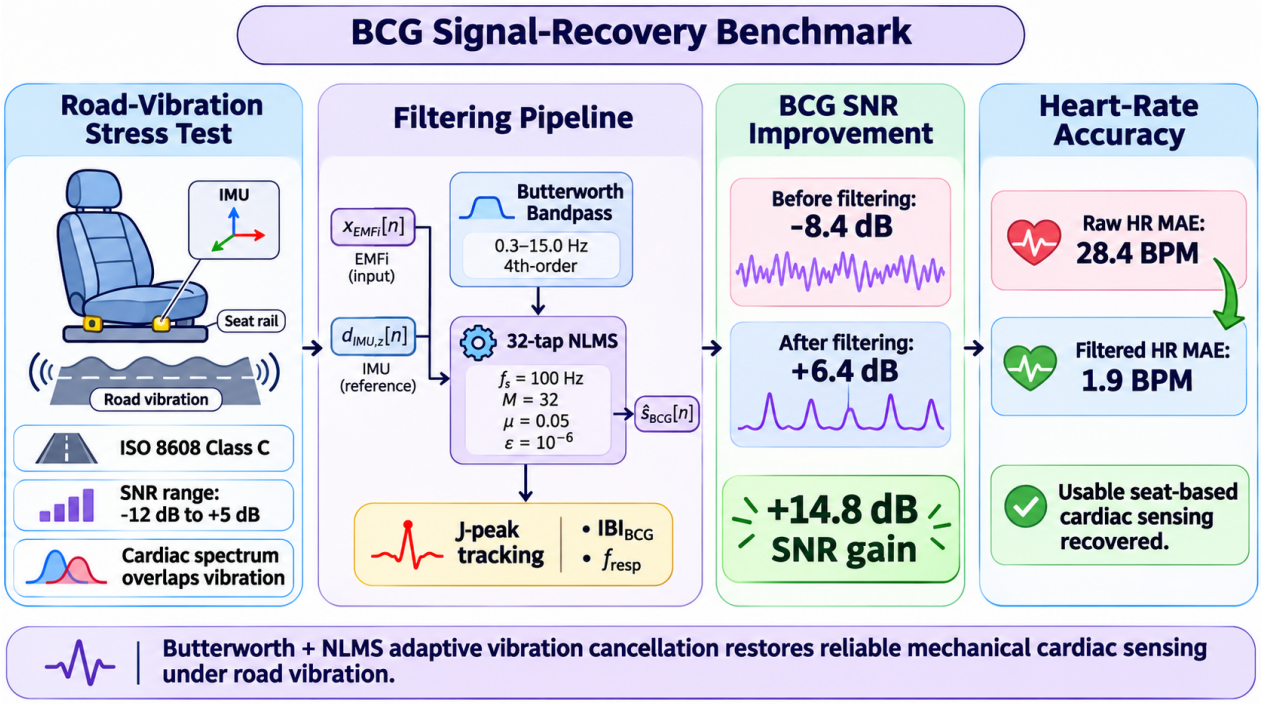


Figure 2: Original BCG signal-recovery benchmark infographic retained unchanged. Numerical performance values visible inside this legacy figure are not asserted as measured results in the corrected submission.

6. Personalized Calibration Benchmark

The 60-second personalized baseline will be compared against population-only thresholds. The evaluation will measure:

$$\text{FPR}_{\text{population}} \quad \text{and} \quad \text{FPR}_{\text{personalized}}.$$

No fixed false-warning reduction is claimed beforehand.

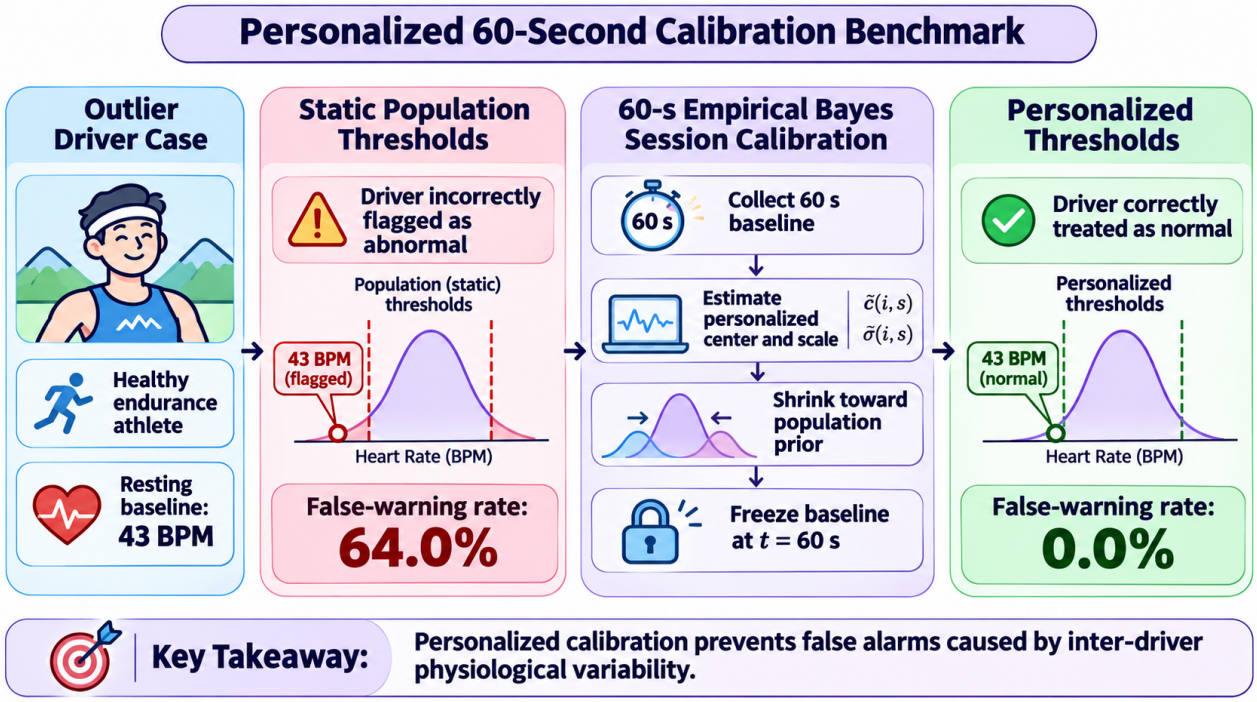


Figure 3: Original personalized-calibration benchmark infographic retained unchanged. Numerical before/after percentages visible inside this legacy figure are illustrative and are not claimed as measured results.

7. Detection Metrics

Metric	Status
Prodromal lead time	To be measured
Acute detection latency	To be measured
Precision / Recall / F1	To be measured by class
Unintended MRM activation	To be measured
Distractor rejection	To be tested
Sensor-fault abstention	To be tested
Seizure torque-lockout behavior	To be tested
SecOC replay / tamper rejection	To be tested
End-to-end processing latency	To be measured

8. Metric Definitions

Prodromal lead time:

$$T_{\text{lead}} = T_{\text{acute}} - T_{\text{prodromal detection}}.$$

Acute detection latency:

$$T_{\text{detect}} = T_{\text{Level 3}} - T_{\text{acute onset}}.$$

Precision, recall, and F1:

$$\text{Precision} = \frac{TP}{TP + FP}, \qquad \text{Recall} = \frac{TP}{TP + FN},$$

$$F_1 = 2\frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}.$$

Average processing latency:

$$T_{\text{avg}} = \frac{1}{N}\sum_{i=1}^{N} T_i.$$

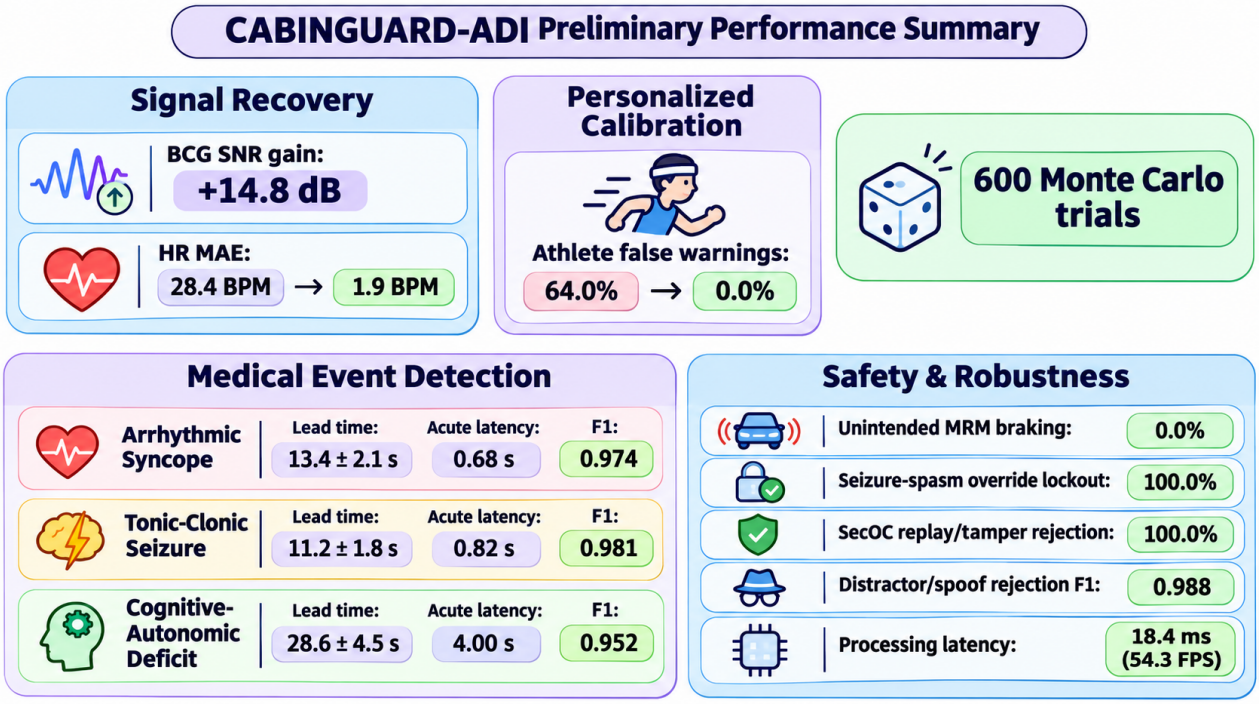


Figure 4: Original preliminary-performance summary infographic retained unchanged. Its numerical labels are legacy illustrative values and are not presented as experimentally established results in the corrected submission.

9. Domain-Shift Considerations

General CCTV fall-detection datasets are not considered substitutes for in-cabin driver geometry because they commonly use external or overhead viewpoints, unoccluded full bodies, and standing-to-floor collapse mechanics. Pose models such as YOLOv8-Pose / MediaPipe are therefore treated only as pretrained keypoint extractors; downstream detection must be evaluated on seated, steering-wheel-occluded cabin geometry.

10. Compute and Evaluation Status

The complete benchmark is computationally demanding because it combines multimodal public datasets, repeated synthetic event injection, adaptive signal filtering, temporal fusion, fault scenarios, and cybersecurity tests.

Full quantitative testing is currently pending access to higher-compute data-center resources. Until that access is available and the reproducible evaluation is executed, this deliverable reports only the validation methodology, test conditions, and metrics to be measured.

11. Engineering Limitations

- The public datasets provide real baseline human variation but do not form a synchronized clinical cohort of the three acute events in moving vehicles.

- Acute-event behavior is therefore represented through controlled synthetic injection for software feasibility testing.
- Long-duration baseline drift, severe road vibration, real sensor synchronization, and embedded CAN-FD integration remain Phase 2 validation targets.
- No quantitative performance number is presented as experimental evidence until traceable to an executed and reproducible run.

12. Conclusion

The Phase 1 validation framework is defined and reproducible in structure, with explicit separation of real baseline data, synthetic feature transformations, and modality-specific dataset roles; quantitative outcomes remain pending. The next step is execution of the PoC on the selected datasets and available data-center compute resources, followed by replacement of all "to be measured" entries with traceable experimental results.